

# Periodic Table of Elements

|                   |                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
|-------------------|-------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| 1<br>IA<br>1A     | 2<br>IIA<br>2A    |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |                    |                    |                    |                    |                    |                    | 13<br>IIIA<br>3A   | 14<br>IVA<br>4A    | 15<br>VA<br>5A     | 16<br>VIA<br>6A    | 17<br>VIIA<br>7A   | 18<br>VIIIA<br>8A  |
| 1<br>H<br>1.008   |                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    | 2<br>He<br>4.003   |
| 3<br>Li<br>6.941  | 4<br>Be<br>9.012  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| 11<br>Na<br>22.99 | 12<br>Mg<br>24.31 | 3<br>IIIB<br>3B   | 4<br>IVB<br>4B                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5<br>VB<br>5B      | 6<br>VIB<br>6B     | 7<br>VIIB<br>7B    | 8<br>VIIIB<br>8B   | 9<br>VIIIB<br>8B   | 10<br>VIIIB<br>8B  | 11<br>IB<br>1B     | 12<br>IIB<br>2B    | 13<br>Al<br>26.98  | 14<br>Si<br>28.09  | 15<br>P<br>30.97   | 16<br>S<br>32.06   | 17<br>Cl<br>35.45  | 18<br>Ar<br>39.95  |
| 19<br>K<br>39.10  | 20<br>Ca<br>40.08 | 21<br>Sc<br>44.96 | 22<br>Ti<br>47.87                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 23<br>V<br>50.94   | 24<br>Cr<br>52.00  | 25<br>Mn<br>54.94  | 26<br>Fe<br>55.85  | 27<br>Co<br>58.93  | 28<br>Ni<br>58.69  | 29<br>Cu<br>63.55  | 30<br>Zn<br>65.38  | 31<br>Ga<br>69.72  | 32<br>Ge<br>72.63  | 33<br>As<br>74.92  | 34<br>Se<br>78.96  | 35<br>Br<br>79.90  | 36<br>Kr<br>83.80  |
| 37<br>Rb<br>85.47 | 38<br>Sr<br>87.62 | 39<br>Y<br>88.91  | 40<br>Zr<br>91.22                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 41<br>Nb<br>92.91  | 42<br>Mo<br>95.96  | 43<br>Tc<br>(97.9) | 44<br>Ru<br>101.1  | 45<br>Rh<br>102.9  | 46<br>Pd<br>106.4  | 47<br>Ag<br>107.9  | 48<br>Cd<br>112.4  | 49<br>In<br>114.8  | 50<br>Sn<br>118.7  | 51<br>Sb<br>121.8  | 52<br>Te<br>127.6  | 53<br>I<br>126.9   | 54<br>Xe<br>131.3  |
| 55<br>Cs<br>132.9 | 56<br>Ba<br>137.3 | 57<br>La<br>138.9 | 72<br>Hf<br>178.5                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 73<br>Ta<br>180.9  | 74<br>W<br>183.8   | 75<br>Re<br>186.2  | 76<br>Os<br>190.2  | 77<br>Ir<br>192.2  | 78<br>Pt<br>195.1  | 79<br>Au<br>197.0  | 80<br>Hg<br>200.6  | 81<br>Tl<br>204.4  | 82<br>Pb<br>207.2  | 83<br>Bi<br>209.0  | 84<br>Po<br>(209)  | 85<br>At<br>(210)  | 86<br>Rn<br>(222)  |
| 87<br>Fr<br>(223) | 88<br>Ra<br>(226) | 89<br>Ac<br>(227) | 104<br>Rf<br>(265)                                                                                                                                                                                                                                                                                                                                                                                                                                                | 105<br>Db<br>(268) | 106<br>Sg<br>(271) | 107<br>Bh<br>(272) | 108<br>Hs<br>(277) | 109<br>Mt<br>(276) | 110<br>Ds<br>(281) | 111<br>Rg<br>(280) | 112<br>Cn<br>(285) | 113<br>Nh<br>(284) | 114<br>Fl<br>(289) | 115<br>Mc<br>(288) | 116<br>Lv<br>(293) | 117<br>Ts<br>(294) | 118<br>Og<br>(294) |
|                   |                   |                   | <div> <div>12<br/>C<br/>12.01</div> <div> <div>Atomic number</div> <div>Atomic symbol</div> <div>Atomic mass</div> </div> </div>                                                                                                                                                                                                                                                                                                                                  |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
|                   |                   |                   | <div> <div>58<br/>Ce<br/>140.1</div> <div>59<br/>Pr<br/>140.9</div> <div>60<br/>Nd<br/>144.2</div> <div>61<br/>Pm<br/>(145)</div> <div>62<br/>Sm<br/>150.4</div> <div>63<br/>Eu<br/>152.0</div> <div>64<br/>Gd<br/>157.3</div> <div>65<br/>Tb<br/>158.9</div> <div>66<br/>Dy<br/>162.5</div> <div>67<br/>Ho<br/>164.9</div> <div>68<br/>Er<br/>167.3</div> <div>69<br/>Tm<br/>168.9</div> <div>70<br/>Yb<br/>173.0</div> <div>71<br/>Lu<br/>175.0</div> </div>    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
|                   |                   |                   | <div> <div>90<br/>Th<br/>232.0</div> <div>91<br/>Pa<br/>231.0</div> <div>92<br/>U<br/>238.0</div> <div>93<br/>Np<br/>(237)</div> <div>94<br/>Pu<br/>(244)</div> <div>95<br/>Am<br/>(243)</div> <div>96<br/>Cm<br/>(247)</div> <div>97<br/>Bk<br/>(247)</div> <div>98<br/>Cf<br/>(251)</div> <div>99<br/>Es<br/>(252)</div> <div>100<br/>Fm<br/>(257)</div> <div>101<br/>Md<br/>(258)</div> <div>102<br/>No<br/>(259)</div> <div>103<br/>Lr<br/>(262)</div> </div> |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |

## Physical Constants

| Name                    | Symbol       | Value                                                                                          |
|-------------------------|--------------|------------------------------------------------------------------------------------------------|
| Avogadro's Number       | $N$          | $6.0221 \times 10^{23} \text{ mol}^{-1}$                                                       |
| Boltzmann's Constant    | $k_B$        | $1.3806 \times 10^{-23} \text{ J K}^{-1}$                                                      |
| Electron Charge         | $e$          | $1.6022 \times 10^{-19} \text{ C}$                                                             |
| Faraday's Constant      | $F = Ne$     | $9.6485 \times 10^4 \text{ C mol electrons}^{-1}$                                              |
| Ideal Gas Constant      | $R = Nk_B$   | $0.082057 \text{ L atm K}^{-1} \text{ mol}^{-1}$<br>$8.3145 \text{ J K}^{-1} \text{ mol}^{-1}$ |
| Planck's Constant       | $h$          | $6.6261 \times 10^{-34} \text{ J s}$                                                           |
| Rydberg Constant        | $R_H$        | $1.0968 \times 10^7 \text{ m}^{-1}$                                                            |
|                         | $hcR_H$      | $2.1787 \times 10^{-18} \text{ J}$                                                             |
| Speed of light (vacuum) | $c$          | $2.9979 \times 10^8 \text{ m s}^{-1}$                                                          |
| Vacuum permittivity     | $\epsilon_o$ | $8.8542 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$                             |
| Mass of electron        | $m_e$        | $9.1094 \times 10^{-31} \text{ kg}$                                                            |
| Mass of neutron         | $m_n$        | $1.6749 \times 10^{-27} \text{ kg}$                                                            |
| Mass of proton          | $m_p$        | $1.6726 \times 10^{-27} \text{ kg}$                                                            |

## Conversion Factors

|                                     |
|-------------------------------------|
| <b>Distance</b>                     |
| 2.54 cm = 1 in                      |
| <b>Volume</b>                       |
| 1 cm <sup>3</sup> = 1 mL            |
| <b>Energy</b>                       |
| 1 cal = 4.184 J                     |
| 1 Cal = 1 kcal                      |
| 1 J = 1 m <sup>3</sup> Pa           |
| <b>Pressure</b>                     |
| 1 atm = 101325 Pa                   |
| 1 bar = 1 × 10 <sup>5</sup> Pa      |
| 1 atm = 760 Torr                    |
| 1 mmHg = 1 Torr                     |
| <b>Dipole moment</b>                |
| 1 D = 3.336 × 10 <sup>-30</sup> C m |